

**FINAL EXAMINATION
PYTHON FOR DATA SCIENCE**

EXAM VERSION: ODD

Full name: _____ **Student ID:** _____
Class: _____ **Room:** _____
Signature: _____ **Exam date:** _____

Duration: 120 minutes (2 hours). Total: 10.0 points.
The exam has **three parts**: (I) Multiple Choice - 3.0 points; (II) Logical Reasoning and Short Answers - 3.0 points; (III) Integrated Data Problems - 4.0 points.

INSTRUCTIONS

1. Write **directly on this exam paper** and keep each answer inside the designated answer area.
2. For output-prediction questions, reason from the code; you do not need to run Python.
3. For coding questions, write only the necessary statements; do not copy the given data again.
4. Equivalent correct solutions are accepted when the logic and result are correct.
5. Show clear reasoning where requested. Partial credit may be awarded for a correct method even if a final arithmetic value is slightly inaccurate.

Part	I	II	III	Total
Maximum	3.0	3.0	4.0	10.0
Score				

PART I - MULTIPLE CHOICE (20 questions x 0.15 = 3.0 points)

Select **one** best answer for each question and mark A/B/C/D directly on this paper.

1. What is the result of "7" * 3?

A. 21 B. "21" C. "777" D. Error

Answer: ☐ A ☐ B ☐ C ☐ D

2. Given label = "AI-NEU-2026", what is label[3:6]?

A. AI- B. NEU C. EU- D. NEU-

Answer: ☐ A ☐ B ☐ C ☐ D

3. Since Python strings are immutable, which is the most appropriate way to change "data" to "Data"?

A. Assign s[0]="D"

B. Reassign s from an expression that creates a new string

C. Use append

D. Use insert

Answer: ☐ A ☐ B ☐ C ☐ D

4. To validate an entire code consisting of SV followed by exactly 6 digits, use:

A. re.fullmatch(r"SV\d{6}", value)

B. re.search(r"SV", value)

C. re.findall(r"\d+", value)

D. re.split(r"SV", value)

Answer: ☐ A ☐ B ☐ C ☐ D

5. Which statement replaces one or more whitespace characters with one space and trims the ends?

A. re.sub(r"\s+", " ", text).strip()

B. re.search(r"\s+", text)

C. re.findall(r"\s+", text)

D. text.index(" ")

Answer: ☐ A ☐ B ☐ C ☐ D

6. np.linspace(0, 1, 5) creates:

A. 5 evenly spaced values from 0 to 1, including both endpoints

B. [0,1,2,3,4]

C. 5 random integers

D. [0,0.5,1]

Answer: ☐ A ☐ B ☐ C ☐ D

7. For B=np.array([[1,2],[3,4],[5,6]]), B.shape is:

A. (2,3) B. (3,2) C. (6,) D. 6

Answer: ☐ A ☐ B ☐ C ☐ D

8. For the array in Question 7, B[1:,0] is:

A. [1,3,5] B. [3,5] C. [4,6] D. [3,4]

Answer: ☐ A ☐ B ☐ C ☐ D

9. If c.shape==(2,1) and A.shape==(2,3), then A+c will:

A. Add one element of c to each corresponding row

B. Always raise an error

C. Add only to the first column

D. Transpose A

Answer: ☐ A ☐ B ☐ C ☐ D

10. For a 2-D array, A.mean(axis=1) normally returns:

A. One mean per row

B. One mean per column

C. Always one scalar

D. The number of elements

Answer: ☐ A ☐ B ☐ C ☐ D

PART I - MULTIPLE CHOICE (continued)

11. A Pandas DataFrame is best described as:

- A. A one-dimensional labeled array
- B. A two-dimensional table of rows and columns
- C. Only a Python list
- D. A structure that stores numeric data only

Answer: ☐ A ☐ B ☐ C ☐ D

12. To select the first row by **integer position**, use:

- A. `df.loc[0]` B. `df.iloc[0]`
- C. `df[0]` D. `df.first()`

Answer: ☐ A ☐ B ☐ C ☐ D

13. Which correctly filters rows where `Region=="North"` **or** `Revenue>1000`?

- A. `df[(df.Region=="North") | (df.Revenue>1000)]`
- B. `df[df.Region=="North" or df.Revenue>1000]`
- C. `df[(df.Region=="North") & (df.Revenue>1000)]`
- D. `df.loc["North",1000]`

Answer: ☐ A ☐ B ☐ C ☐ D

14. To count rows for each Category, use:

- A. `df["Category"].value_counts()`
- B. `df["Category"].mean()`
- C. `df.sort_values("Category")`
- D. `df.describe("Category")`

Answer: ☐ A ☐ B ☐ C ☐ D

15. Which function directly removes duplicate rows?

- A. `dropna()` B. `drop_duplicates()`
- C. `fillna()` D. `duplicated().mean()`

Answer: ☐ A ☐ B ☐ C ☐ D

16. For a very large CSV that should be processed in parts, which parameter is important?

- A. `chunksize` B. `index=False`
- C. `ascending` D. `axis=1`

Answer: ☐ A ☐ B ☐ C ☐ D

17. JSON Lines is especially suitable for:

- A. Raster images
- B. Record-oriented web/API data, one record per line
- C. Only dense numeric matrices
- D. Replacing every database system

Answer: ☐ A ☐ B ☐ C ☐ D

18. After loading a dataset, which check is most directly useful for identifying the number of missing values in each column?

- A. `df.isna().sum()`
- B. `df.columns.mean()`
- C. `df.sort_values()`
- D. `df.shape.sum()`

Answer: ☐ A ☐ B ☐ C ☐ D

19. To inspect the shape of the distribution of one continuous numeric variable, prefer a:

- A. Histogram B. Pie chart
- C. Network graph D. Single table only

Answer: ☐ A ☐ B ☐ C ☐ D

20. Truncating the Y-axis close to the observed values can:

- A. Always make a chart more accurate
- B. Make small differences look much larger and potentially mislead the reader
- C. Remove outliers automatically
- D. Change the underlying data

Answer: ☐ A ☐ B ☐ C ☐ D

PART II - LOGICAL REASONING AND SHORT ANSWERS (3.0 points)**Question 21.****[0.75 points]**

Given:

```
B = np.array([
    [3, 6, 9],
    [6, 9, 12],
    [9, 12, 15],
    [12, 15, 18]
])
```

a) Determine `B.shape` and `B.size`.*Write your answer in this box.***b)** Write the result of `B[1:3, 0:2]`.*Write your answer in this box.***c)** Write the result of `B[:, :2, :]`.*Write your answer in this box.***d)** Calculate `B.mean(axis=1)` and briefly explain what `axis=1` means here.*Write your answer in this box.***e)** Given `x=np.array([2,9,14,5,11,18,7])`, write an expression selecting values **from 7 through 14, inclusive**; also write the resulting values.*Write your answer in this box.*

PART II - LOGICAL REASONING AND SHORT ANSWERS (continued)**Question 22.****[0.75 points]**

Consider the following DataFrame:

OrderID	Category	Quantity	UnitPrice	Region
O01	Tech	2	500	North
O02	Book	5	80	South
O03	Tech	1	1200	North
O04	Home	3	300	Central
O05	Book	10	60	South
O06	Home	2	450	North

Assume the table is stored in df. Write the necessary Pandas statements.

a) Filter orders in North with UnitPrice >= 500.*Write your answer in this box.***b)** Create Revenue = Quantity * UnitPrice.*Write your answer in this box.***c)** Compute total Revenue by Category.*Write your answer in this box.***d)** Identify the problem in the following code and correct it.

```
df[df["Quantity"] >= 2 and df["UnitPrice"] > 300]
```

*Write your answer in this box.***e)** Write one expression returning the OrderID of the row with the highest Revenue.*Write your answer in this box.*

PART II - LOGICAL REASONING AND SHORT ANSWERS (final page)**Question 23.****[0.75 points]**

Given:

```
text = "Order: OD-2048 | Email: lan@shop.vn | Phone: 0987 654 321"
```

a) Write a Regex pattern that extracts an order code of OD- followed by exactly 4 digits.*Write your answer in this box.***b)** Write one `re.sub` statement that normalizes the phone number so that only digits remain.*Write your answer in this box.***c)** Briefly explain the difference in purpose between `findall()` and `fullmatch()`.*Write your answer in this box.***Question 24.****[0.75 points]**

The goal is to read `sales.csv`, keep rows with `Revenue > 1000`, print the **mean Revenue by Region**, and save exactly the filtered rows to `high_sales.csv` without an extra index column.

```
sales = pd.read_csv("sales.csv")
high = sales[sales["Revenue"] > 1000]
print(high.groupby("Region")["Revenue"])
sales.to_csv("high_sales.csv")
```

Identify and correct all incomplete or incorrect statements.

Write your answer in this box.

PART III - INTEGRATED DATA PROBLEMS (4.0 points)**Question 25.****[2.0 points]****Sales data analysis.** Given:

OrderID	Category	Quantity	UnitPrice	Discount	Region
O01	Tech	2	500	0.10	North
O02	Book	5	80	0.00	South
O03	Tech	1	1200	0.05	North
O04	Home	3	300	0.10	Central
O05	Book	10	60	0.05	South
O06	Home	2	450	0.00	North
O07	Tech	4	250	0.10	Central
O08	Home	5	200	0.05	South

Assume the data are stored in df.

a) [0.25] Create `NetRevenue = Quantity*UnitPrice*(1-Discount)`.*Write your answer in this box.***b) [0.25]** Without running Python, identify the OrderID with the highest NetRevenue and give that value. Show enough calculation to justify the answer.*Write your answer in this box.***c) [0.30]** Write code to filter orders with `Quantity >= 2` and `NetRevenue >= 900`.*Write your answer in this box.***d) [0.30]** Write code to compute mean NetRevenue by Category.*Write your answer in this box.*

QUESTION 25 - CONTINUED

e) [0.30] Which chart would you choose to investigate the relationship between UnitPrice and NetRevenue? Give one reason.

Write your answer in this box.

f) [0.30] Write code to keep exactly OrderID, Category, NetRevenue and sort rows by NetRevenue from highest to lowest.

Write your answer in this box.

g) [0.30] The processed table will be reused for analysis on a large dataset and should preserve data types well. Choose among **CSV**, **JSON Lines**, or **Parquet** and justify your choice.

Write your answer in this box.

Reasoning reminder: choose a storage format based on the intended analysis workflow, data size, read efficiency, interoperability, and data-type preservation - not only on whether the file is human-readable.

QUESTION 26 - DATA RETRIEVING, STORING, AND VISUALIZATION**Question 26.****[2.0 points]**

A file `orders.csv` has **18 million rows** and the columns:

OrderID, Date, Region, Channel, Quantity, Revenue, Returned, CustomerID, Note

You need to build a sales-analysis pipeline.

a) [0.25] Only OrderID, Date, Region, Revenue are needed. Write a `read_csv` statement that reads only these columns and parses Date as datetime.

Write your answer in this box.

b) [0.25] The machine cannot load all 18 million rows at once. Write a chunked-reading approach using, for example, 200,000 rows per chunk.

Write your answer in this box.

c) [0.25] Some Revenue values are missing because orders are still pending confirmation. Explain why blindly applying `dropna()` to the whole dataset may be inappropriate, and propose a more defensible treatment.

Write your answer in this box.

d) [0.25] After processing, which format would you choose for a large analytical table that should reload quickly and preserve data types reasonably well? Explain why.

Write your answer in this box.

QUESTION 26 - CONTINUED

Assume the cleaned data are in a DataFrame named `orders` with columns `OrderID`, `Date`, `Region`, `Channel`, `Quantity`, `Revenue`, `Returned`.

e) [0.35] Write Pandas code that keeps only rows with `Returned == 0` and computes both mean `Quantity` and mean `Revenue` for each `Channel`.

Write your answer in this box.

f) [0.25] To show **average Revenue by day** across 30 consecutive days, which chart would you choose? Give a short reason.

Write your answer in this box.

g) [0.20] A scatter plot shows a positive association between `Quantity` and `Revenue`. A student concludes: “Buying a larger quantity definitely causes revenue to increase.” Explain the logical problem with this conclusion.

Write your answer in this box.

h) [0.20] After a `Store → Reload` cycle, state **two checks** that help verify that the data were stored and reloaded correctly.

Write your answer in this box.

— END OF EXAM —